

IS547 Course Final Project Report
**Harmonizing Instrumental Climate Observations and Reported
Severe Weather Events in Illinois (2010–2025)**

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GitHub Repository: https://github.com/jazzduvv-56/IS547_Course_Project

Zenodo Sandbox Record: <https://sandbox.zenodo.org/records/495348>

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Abstract

This project curates an integrated dataset that aligns daily instrumental weather measurements from the NOAA Global Historical Climatology Network - Daily (GHCN-Daily) with county-level severe weather event records from the NOAA Storm Events Database for the state of Illinois between 2010 and 2025. The main curation goal was to examine how these two authoritative, but structurally different NOAA systems represent the same physical weather events. I selected data from ten Illinois weather stations and harmonized it with four types of severe weather events using a custom station-to-county lookup table.

Following the DCC Curation Lifecycle Model (Higgins, 2008), I automated the entire process from acquisition and cleaning to spatial and temporal harmonization using Python scripts with relative paths. I created a detailed data dictionary, comprehensive README, and rich metadata, and deposited the final integrated dataset in Zenodo Sandbox (DOI: 10.5072/zenodo.495348) under a Creative Commons Attribution 4.0 license. This work demonstrates practical application of key data curation concepts including data integration, metadata quality, provenance tracking, and reproducibility. The resulting dataset provides a transparent, reusable resource for researchers studying the alignment between measured weather conditions and officially reported severe weather events in Illinois.

Introduction and Motivation

As a master's student, I often think about how researchers need clean, well-aligned data to answer real questions. For this project, I took on the role of a data curator working with environmental researchers in Illinois. I wanted to create a dataset that would help them study how measured weather conditions relate to officially reported severe weather events.

The main curation goal was to harmonize daily instrumental measurements from the NOAA Global Historical Climatology Network - Daily (GHCN-Daily) with county-level severe weather event records from the NOAA Storm Events Database for the years 2010 to 2025. Instead of trying to prove that snow causes blizzards or high temperatures cause heat warnings, I focused on a more fundamental curation question: how do these two authoritative NOAA systems represent the same physical weather events differently? This includes differences in spatial scale, temporal detail, and classification rules.

I deliberately kept the scope manageable by selecting ten weather stations spread across Illinois and focusing on four key severe weather event types: Winter Storm, Blizzard, Flood, and Excessive Heat. The time range of 2010 to 2025 gave me recent data while keeping the volume reasonable for a semester project. These choices let me concentrate on the real curation challenges, such as mapping station locations to counties and aligning daily records with multi-day events.

This project was especially meaningful to me because it gave me hands-on experience with the kind of data cleaning, integration, and harmonization tasks I hope to do in a data engineering role after graduation.

The prioritized curation goals for this project were: (1) improving fitness-for-use through spatial and temporal harmonization, (2) ensuring transparency by documenting all integration decisions, (3) maximizing reusability with clear metadata and a data dictionary, and (4) achieving full reproducibility through relative paths and a public GitHub repository. These goals directly led to the data dictionary, comprehensive README, reproducible Python pipeline, and Zenodo deposit in the final dataset.

Datasets and Context

I worked with two well-known NOAA datasets for this project. The first is the Global Historical Climatology Network Daily (GHCN-Daily) (Menne et al., 2012). This dataset provides daily instrumental measurements from thousands of weather stations worldwide. For my work, I used data from ten stations spread across Illinois. The main variables I kept were daily maximum and minimum temperature, average temperature (calculated), precipitation, snowfall, and snow depth.

The second dataset is the NOAA Storm Events Database. Unlike GHCN-Daily, this one records officially reported severe weather events at the county level. It includes details such as event type, begin and end dates, property damage, crop damage, injuries, and deaths. I focused on four event types that are common in Illinois: Winter Storm, Blizzard, Flood, and Excessive Heat.

These two datasets describe the same real-world weather but in very different ways. GHCN-Daily gives precise daily measurements at specific locations, while Storm Events gives broader county-level summaries of events that may last multiple days. This difference in structure, spatial scale, and classification created the main curation challenges I had to solve. Both datasets are public domain and come from trusted government sources, which made them suitable for reuse. I downloaded the raw files directly from the official NOAA NCEI websites (National Centers for Environmental Information, n.d.-a, n.d.-b).

I chose these specific datasets because they are widely used, well-documented, and directly relevant to studying how measured conditions relate to officially reported events in Illinois. The time range was limited to 2010–2025 to keep the volume manageable while still covering recent years.

Ethical, Legal, and Policy Considerations

Both datasets I used are produced by the U.S. government and are in the public domain. NOAA makes GHCN-Daily and the Storm Events Database freely available with no copyright restrictions,

so there were no licensing issues or legal barriers to downloading, cleaning, or integrating the data. I also did not have to worry about privacy concerns because neither dataset contains any personal or identifiable information.

Even though the data is openly available, I still thought carefully about ethical responsibilities. The main ethical issue for me was making sure the integrated dataset does not lead to misinterpretation. For example, just because a day has high snowfall does not automatically mean it was officially recorded as a “Blizzard.” By clearly documenting the differences between measured conditions and reported events, I tried to reduce the risk that someone might draw incorrect conclusions from the combined data. These considerations directly affect the dataset’s fitness-for-use, particularly when researchers want to draw causal conclusions from measured conditions and reported events.

I also made sure to give proper credit to the original sources by linking back to the NOAA records in the metadata and README. Overall, working with public government data made the ethical and legal side straightforward, but it still reminded me how important it is to be transparent about the limitations and decisions I made during curation.

Data Curation Workflow

I structured the entire project around the DCC Curation Lifecycle Model (Higgins, 2008), which we studied in class. This model helped me think systematically about each stage from planning through dissemination.

The workflow began with acquisition. I downloaded the raw GHCN-Daily files for the ten selected Illinois stations and the Storm Events Database records for Illinois directly from the official NOAA NCEI websites. Next came cleaning and quality assurance. For GHCN-Daily, I used pandas to filter the ten stations, converted temperature from Fahrenheit to Celsius and precipitation/snow from inches to millimeters, calculated average temperature where missing, and handled trace values and flags. For the Storm Events data, I filtered to the four relevant event types and expanded multi-day events into daily records using glob and pandas.

The most challenging part was integration and harmonization. Because GHCN-Daily is point-based and Storm Events is county-based, I created a custom station-to-county lookup table and added a `normalize_county` function to handle naming differences. I chose this transparent approach instead of a full shapefile spatial join because it was more reproducible for a semester project. All steps were automated using Python scripts with relative paths only, and I tracked every transformation in the README and data dictionary. The final integrated dataset was then deposited on Zenodo Sandbox to support preservation and dissemination.

This workflow gave me a practical understanding of how the DCC Curation Lifecycle Model works in real curation work, particularly the stages of acquisition, transformation, and reuse.

Metadata, Documentation, and Identifiers

One of the most important parts of this project was creating clear metadata and documentation so an independent user could understand and reuse the dataset. I produced a detailed data dictionary that defines every column in the final integrated file, including variable names, descriptions, units (TMAX/TMIN in °C, PRCP/SNOW in mm), and transformations applied. I also wrote a comprehensive README describing the project purpose, data sources, cleaning steps, and reproduction instructions.

I completed as many Zenodo metadata fields as possible, including keywords, temporal coverage (2010–2025), and related works linking back to the original NOAA sources. These efforts directly support FAIR principles by improving findability and reusability (Wilkinson et al., 2016).

For identifiers, I used official GHCN station codes and created a unique record identifier combining station and date, while the Zenodo DOI provides a persistent identifier for the curated dataset. These efforts were aimed at making the dataset truly discoverable and citable, something I learned is often missing in raw government data.

Archiving and Preservation

I deposited the final integrated dataset in Zenodo Sandbox (DOI: 10.5072/zenodo.495348) under a Creative Commons Attribution 4.0 license. I chose Zenodo because it provides persistent DOIs, automatic replication across multiple locations to guard against bit rot, and support for format migration, concepts we discussed in class regarding the OAIS model and trustworthy digital repositories. The dataset is saved in open, non-proprietary CSV format, which is widely considered preservation-friendly. Together with the public GitHub repository, this creates a practical two-tier strategy: immediate access on GitHub and stable long-term archiving on Zenodo.

Provenance and Reproducibility

Provenance and reproducibility were important focuses for me throughout the project. I made sure every step was traceable so that someone else could understand exactly how the final dataset was created.

I downloaded the raw GHCN-Daily and Storm Events files directly from the official NOAA NCEI websites on specific dates (documented in the README). All cleaning and integration steps were written in Python scripts using only relative paths and common libraries (pandas, glob). I kept the intermediate files so that anyone can see how the data changed from raw to cleaned to the final integrated version.

The entire workflow is now available in a public GitHub repository. Anyone can clone the repo, place the original files in the “Raw Data” folder, and run the notebook in order to reproduce the exact same final dataset. I also included a simple station-to-county lookup table so the spatial mapping step is completely transparent. These efforts helped me put into practice the provenance and reproducibility concepts we covered in class (Higgins, 2008).

Findings, Challenges, and Tradeoffs

The final integrated dataset contains 52,783 station-days from ten Illinois weather stations between 2010 and 2025. Of these, 1,093 days (approximately 2%) had a matching severe weather event recorded in the NOAA Storm Events Database. The most common events were Flood (590 days), Excessive Heat (265 days), Winter Storm (208 days), and Blizzard (30 days). There were also 1,699 days with measurable snowfall.

The visualizations reveal several meaningful patterns. Temperature histograms (see Figure 1) show that Winter Storm and Blizzard days occurred on significantly colder days than non-event days, while Excessive Heat events were concentrated on the hottest days. This demonstrates successful alignment between instrumental measurements and official event reports. A stacked bar chart of events by season (see Figure 2) further confirms this seasonal alignment: summer clearly dominates with Flood and Excessive Heat events, while winter events are concentrated in the colder months, as expected for Illinois climate.

The yearly trend (see Figure 3) shows considerable year-to-year variation, with a noticeable peak in flood-related events around 2019. The boxplot of daily precipitation (see Figure 4) demonstrates clear separation between Flood event days and non-event days, with flood days having much higher precipitation values. Finally, the county-level match rate (see Figure 5) highlights spatial differences: Rock Island and St. Clair counties had the highest percentage of days with reported events, while Macon County had the lowest.

Several challenges emerged during curation. The biggest was harmonizing the different spatial scales. GHCN-Daily data are point measurements at weather stations, while Storm Events data are reported at the county level. This required building a manual station-to-county lookup table. The approach worked well overall, but it introduced a known limitation: Cook County has two stations (O’Hare and Midway), so the same county-level storm events are matched to both stations, slightly inflating Cook County counts. I documented this tradeoff rather than attempting a more complex spatial join with county boundary shapefiles, as that would have gone beyond the scope of this project.

A second challenge I encountered was that the Joliet station (USW00014834) had no records for most of the period 2010–2024, which limited the completeness of data for that location.

Another challenge was the difference in how the two systems record phenomena. For example, there are far more days with measurable snowfall (1,699) than official Winter Storm or Blizzard declarations (238 combined). This reflects the inherent difference between continuous instrumental measurement and threshold-based event reporting.

These tradeoffs highlight core data curation issues discussed in class: spatial and semantic heterogeneity, the importance of clear documentation of integration decisions, and the value of making limitations transparent for future users.

Conclusion

This project successfully produced a clean and reproducible integrated dataset that aligns daily instrumental weather measurements from GHCN-Daily with county-level severe weather event records from the Storm Events Database for Illinois (2010–2025). The final dataset enables direct comparison between measured conditions and officially reported events.

The work reinforced several core concepts from the course. Reconciling differences in spatial scale and classification rules required careful harmonization and transparent documentation of decisions. The low match rate between the two systems highlighted the real challenges of data integration, even when working with trusted government sources.

Overall, this hands-on experience showed me that good curation is not just about cleaning files but about making deliberate choices that improve fitness for use. The resulting artifacts, including Python scripts with relative paths, a complete data dictionary, and the Zenodo deposit, provide a transparent and reproducible workflow.

This project was one of the most valuable parts of the course. It directly connected theoretical ideas like the DCC Curation Lifecycle Model, data integration, provenance, and reproducibility to practical outcomes.

GenAI usage declaration:

I used Grok to assist with the preparation of this final project report. Specifically, Grok helped me with organizing the structure and flow of the report, drafting and refining the wording of sections for clarity and readability, improving sentence structure, removing repetitive phrasing, and refining the python code.

All ideas, analysis, curation decisions, and final content are my own. I reviewed and edited every part of the report before submission.

References

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Appendix

Figure 1:

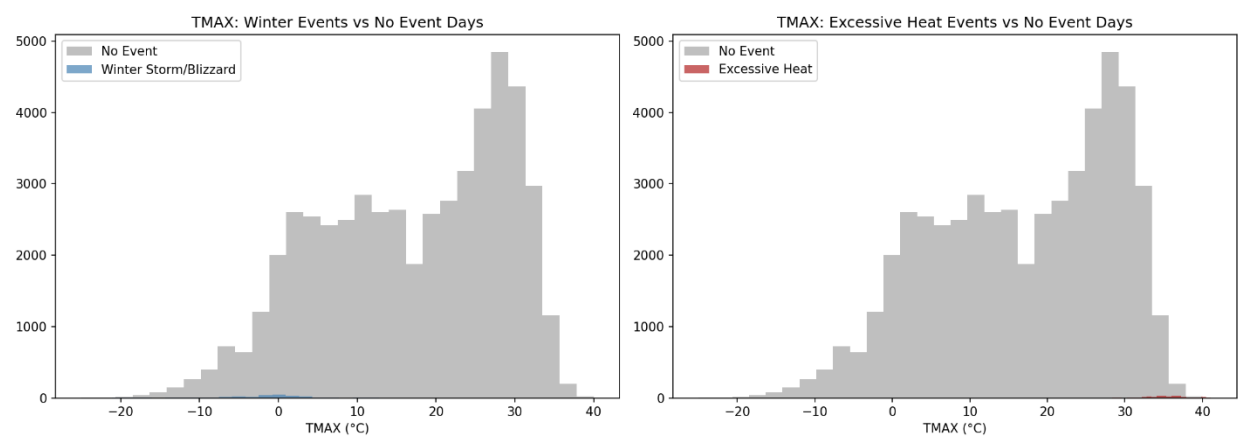


Figure 2:

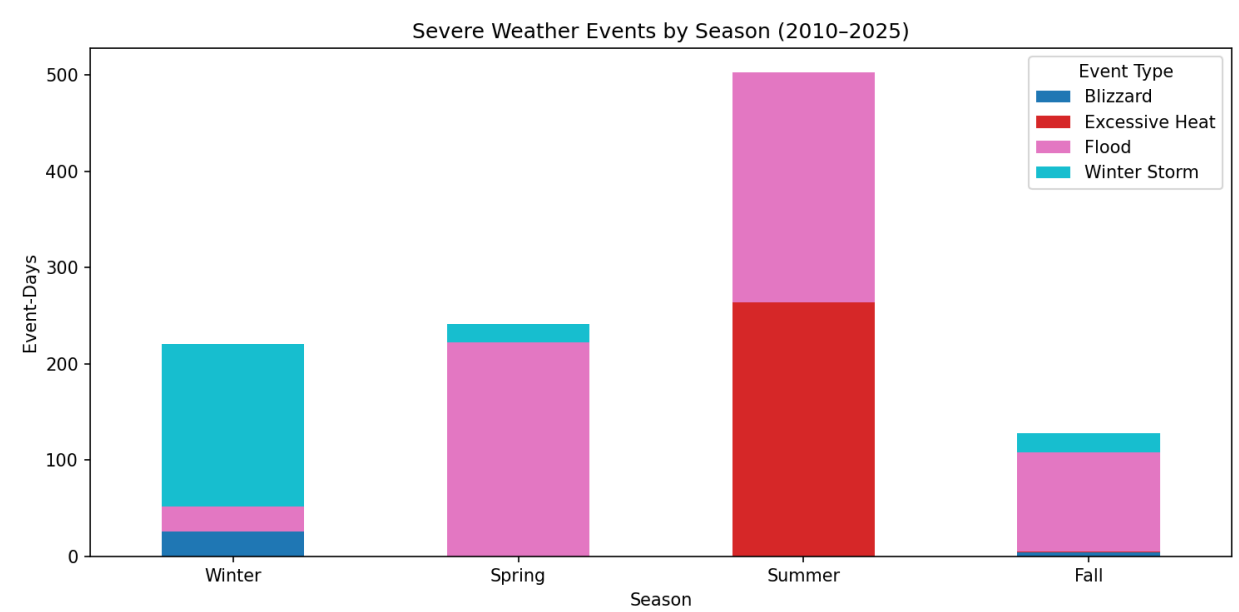


Figure 3:

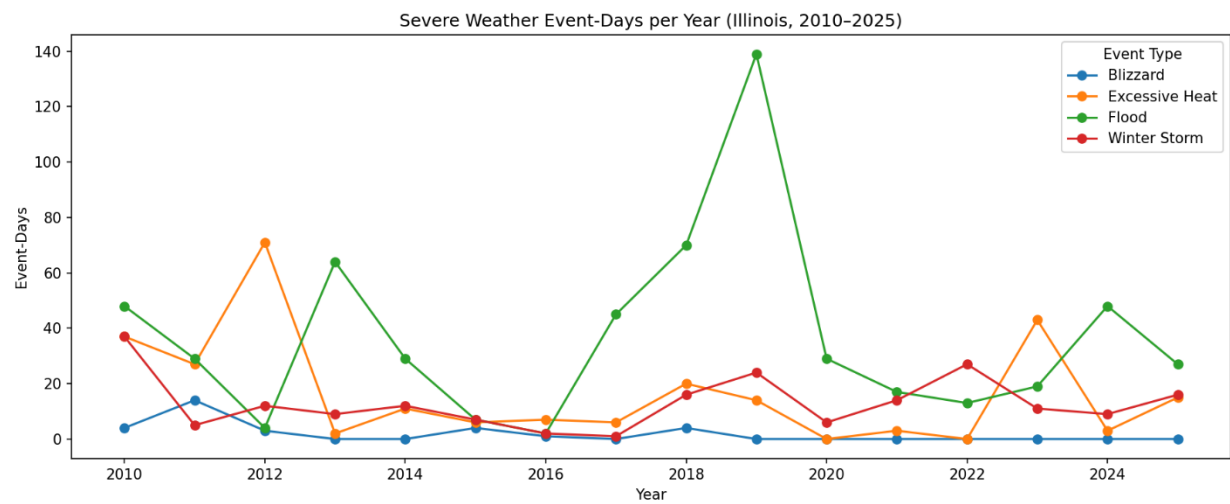


Figure 4:

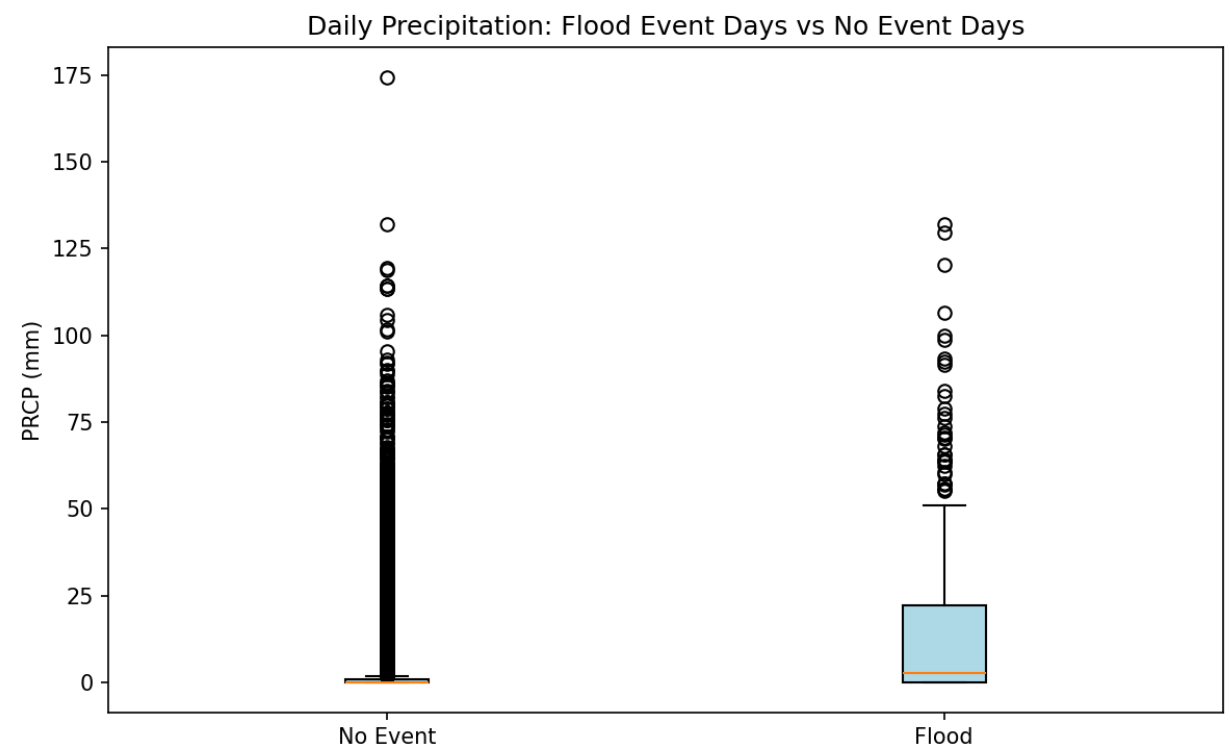


Figure 5:

